

CRN-SF

Installation and operating instructions



English (GB)	
Installation and operating instructions	4
Dansk (DK)	
Monterings- og driftsinstruktion	6
Deutsch (DE)	
Montage- und Betriebsanleitung	8
Ελληνικά (GR)	
Οδηγίες εγκατάστασης και λειτουργίας	10
Español (ES)	
Instrucciones de instalación y funcionamiento	12
Français (FR)	
Notice d'installation et de fonctionnement	14
Italiano (IT)	
Istruzioni di installazione e funzionamento	16
Nederlands (NL)	
Installatie- en bedieningsinstructies	18
Português (PT)	
Instruções de instalação e funcionamento	20
Русский (RU)	
Руководство по монтажу и эксплуатации	22
Suomi (FI)	
Asennus- ja käyttöohjeet	24
Svenska (SE)	
Monterings- och driftsinstruktion	26
Appendix 1	28

CONTENTS

	Page
1. Symbols used in this document	4
2. General information	4
3. Type key	4
4. Applications	4
5. Technical data	4
5.1 Minimum inlet pressure	4
5.2 Maximum inlet pressure	4
5.3 Maximum operating pressure	4
5.4 Dimensions	4
5.5 Electrical data	4
6. Installation	5
6.1 Example	5
7. Start-up	5
8. Disposal	5

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

**Warning**

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General information

This document is a supplement to installation and operating instructions of CR, CRI, CRN standard pumps, publication number 96462123.

3. Type key

Example	CRN- 10 - 21- SF - X - X - X- XXXX
Type range	
Rated flow rate in m ³ /h	
Number of impellers	
Code for pump version	
Code for pipe connection	
Code for materials	
Code for rubber parts	
Code for shaft seal	

4. Applications

CRN-SF pumps are used for pressure boosting and circulation in systems operating with high pressures. The pump requires a minimum inlet pressure of 2 bar and is usually supplied together with a CRN pump which is installed as a feed pump for the CRN-SF pump.

5. Technical data

5.1 Minimum inlet pressure

2 bar at +20 °C.

If the liquid temperature is higher than +20 °C, the inlet pressure must be increased according to *Fig. B* on page 29.

Example

The operating temperature is +80 °C. The total inlet pressure must thus be 2 bar + 0.5 bar = 2.5 bar.

5.2 Maximum inlet pressure

CRN-SF	Static pressure [bar]	Pressure during operation [bar]
3, 5	15	25
10, 15, 20	10	25
32, 45, 64, 90	10	25

5.3 Maximum operating pressure

50 bar.

Caution

The actual inlet pressure + the maximum pump pressure (at no flow) must not exceed 50 bar.

Caution

The pump must not to run against a closed discharge valve.

During start-up

If both pumps start at the same time, it is not necessary to take the inlet pressure to the CRN-SF pump into account.

Caution

If the pumps cannot start at the same time due to the total locked-rotor current, the CRN-SF pump must start first, followed by the feed pump with a delay of maximum three seconds.

5.4 Dimensions

See *Fig. A* on page 28.

5.5 Electrical data

See the motor nameplate.

6. Installation

See the installation and operating instructions for the standard pump.

6.1 Example

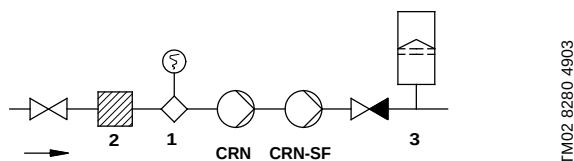


Fig. 1 System with CRN and CRN-SF pump

Pos.	Designation
1	Flow switch
2	Filter
3	Diaphragm tank

The flow switch will protect the pumps against damage caused by overheating by stopping the pumps if the flow rate is too small.

The filter is fitted to protect the flow switch when pumping slightly polluted water.

If the flow rate is stable and higher than the minimum flow rate, the above precautions are not necessary.

If there is a risk of water hammering, a non-return valve should be fitted on the discharge side of the pump.

Caution Do not use rapid-closing valves, e.g. rapid-closing spray guns, as these may cause water hammering.

7. Start-up

See the installation and operating instructions for the standard pump.

Caution Check direction of rotation before starting the pump. If direction of rotation is wrong, the shaft seal will be damaged when the pump is started. Alternatively, you can remove the coupling before checking direction of rotation.

Note The direction of the flow in CRN-SF pumps is the opposite of the one in the standard pump.

System with a feed pump, see section 5.2 Maximum inlet pressure.

8. Disposal

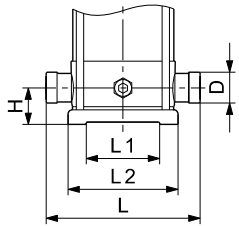
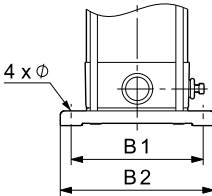
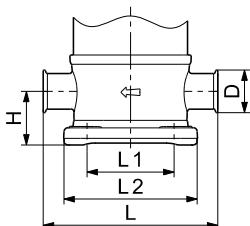
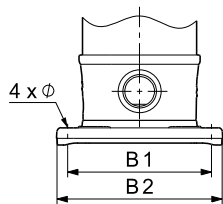
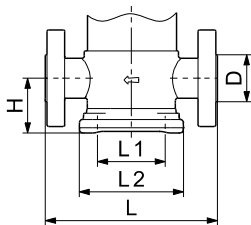
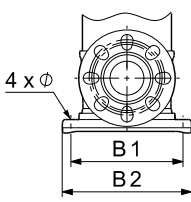
This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Appendix

Fig. A

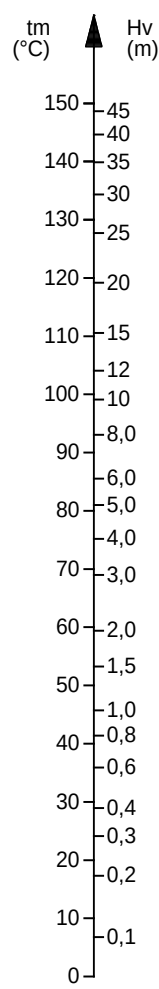
PJE								
Connection type								
Pump type	L [mm]	H [mm]	D [mm]	L₁ [mm]	L₂ [mm]	B₁ [mm]	B₂ [mm]	Ø [mm]
CRN 3 SF CRN 5 SF	210	50	42.2	100	150	180	210	13
CRN 10 SF	261	80	60.1	130	200	215	248	13
CRN 15 SF CRN 20 SF	261	90	60.1	130	200	215	248	13
CRN 32 SF	320	105	88.9	170	226	240	298	14
CRN 45 SF	365	140	114.3	190	251	266	331	14
CRN 64 SF	365	140	114.3	190	251	266	331	14
CRN 90 SF	380	140	114.3	199	261	280	348	14
CX								
Connection type								
Pump type	L [mm]	H [mm]	D [mm]	L₁ [mm]	L₂ [mm]	B₁ [mm]	B₂ [mm]	Ø [mm]
CRN 3 SF CRN 5 SF	210	50	50.5	100	150	180	210	13
CRN 10 SF	261	80	64	130	200	215	248	13
CRN 15 SF CRN 20 SF	261	90	64	130	200	215	248	13
FGJ								
Connection type								
Pump type	L [mm]	H [mm]	D [mm]	L₁ [mm]	L₂ [mm]	B₁ [mm]	B₂ [mm]	Ø [mm]
CRN 3 SF CRN 5 SF	280	85	55	100	150	180	210	13
CRN 10 SF CRN 15 SF CRN 20 SF	330	105	90	130	200	215	248	13

TM02 8371 4309

TM04 5872 4309

TM04 5873 4309

Fig. B



TM00 3037 3493

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana, ramal Campana Centro Industrial Garín - Esq. Haendel y Mozart
AR-1619 Garín Pcia. de Buenos Aires
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 45 3190

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomssesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belorussia

Представительство ГРУНДФОС в Минске
220123, Минск,
ул. В. Хоружей, 22, оф. 1105
Тел.: (+37517) 233 97 65,
Факс: (+37517) 233 97 69
E-mail: grundfos_minsk@mail.ru

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo Branco, 630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztochna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
50/F Maxdo Center No. 8 XingYi Rd.
Hongqiao development Zone
Shanghai 200336
PRC
Phone: +86 21 612 252 22
Telefax: +86 21 612 253 33

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.grundfos.hr

Czech Republic

GRUNDFOS s.r.o.
Čajkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-3066 5650
Telefax: +358-3066 56550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
e-mail: infoservice@grundfos.de
Service in Deutschland:
e-mail: kundendienst@grundfos.de

HILGE GmbH & Co. KG

Hilgestrasse 37-47
55292 Bodenheim/Rhein
Germany
Tel.: +49 6135 75-0
Telefax: +49 6135 1737
e-mail: hilge@hilge.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabaliapuram Road
Thoraiakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg., 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

México

Bombas GRUNDFOS de México S.A. de C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

România

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная 39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47 496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
25 Jalan Tukang
Singapore 619264
Phone: +65-6681 9688
Telefax: +65-6681 9689

Slovenia

GRUNDFOS d.o.o.
Štandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

GRUNDFOS (PTY) LTD
Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentesilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
Box 333 (Lunnagårdsgatan 6)
431 24 Mölndal
Tel.: +46 31 332 23 000
Telefax: +46 31 331 94 60

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloom Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

ТОВ ГРУНДФОС УКРАЇНА
01010 Київ, Вул. Московська 86,
Тел.: (+38 044) 390 40 50
Факс: (+38 044) 390 40 59
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971 4 8815 166
Telefax: +971 4 8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Usbekistan

Представительство ГРУНДФОС в Ташкенте
700000 Ташкент ул.Усмана Носира 1-й тулик 5
Телефон: (3712) 55-68-15
Факс: (3712) 53-36-35

Revised 04.01.2012

96530119 0312
ECM: 1093595

The name Grundfos, the Grundfos logo, and the payoff Be-Think-Innovate are registered trademarks owned by Grundfos Management A/S or Grundfos A/S, Denmark. All rights reserved worldwide.